CocoGuard: material on the models and Grad-CAM

What you can take from the benchmark for your paper and slides: setup, results, figures, suggested wording.

READ THIS FIRST

All numbers here come from the automated benchmark suite executed over **13,836 field images**. This evaluation functions as a **Zero-Shot Cross-Plantation Generalization Test**: the models were trained strictly on the standard 5,798-image dataset, and evaluated across this new 13,836-image collection without any plantation-specific retraining. Use these empirical numbers and figures for your BTP thesis, slides, and technical report.

1. The four models and how they were trained

Metric / Setting	MobileNetV2	ResNet50	EfficientNet-B0	DenseNet-121
ImageNet weights	IMAGENET1K_V1	IMAGENET1K_V1	IMAGENET1K_V1	IMAGENET1K_V1
Parameters (5-class head)	2.23 M	23.52 M	4.01 M	6.96 M
Model Storage Footprint	8.51 MB	89.72 MB	15.31 MB	26.55 MB
Grad-CAM target layer	last feature block	last block of layer4	last feature block	features.denseblock4
Clean accuracy (13,836 imgs)	0.8835	0.8127	**0.9282** Top 1	0.8995
Clean macro-F1 (95% CI)	0.884 [0.879, 0.890]	0.809 [0.801, 0.816]	**0.933 [0.928, 0.937]**	0.905 [0.900, 0.910]
Errors on 13,836 test images	1,612	2,591	**994** (Lowest)	1,390
Clean ECE / Brier score	0.1607 / 0.2147	0.2494 / 0.3579	0.2042 / 0.1757	0.1633 / 0.1899

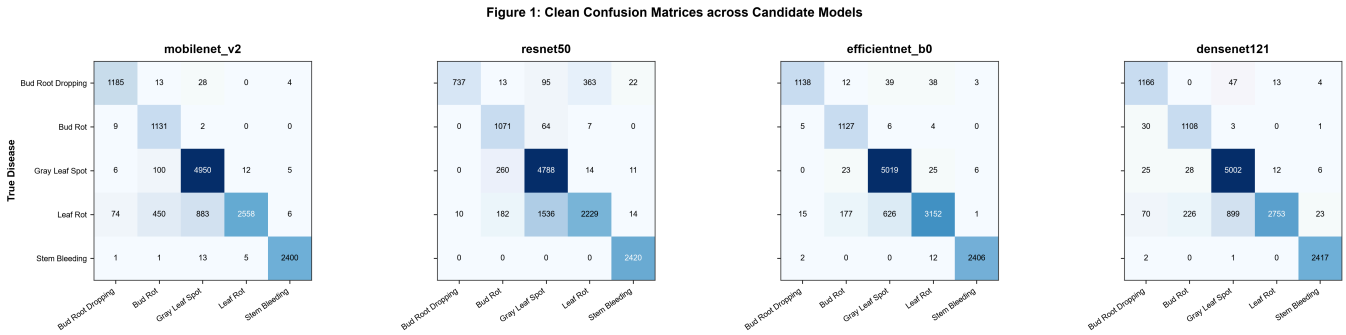


Figure 1. Clean confusion matrices (13,836 test images). *Across all models, Stem Bleeding (trunk pathology) achieved near-perfect accuracy (2,406 / 2,420 correct on EfficientNet-B0). The majority of errors occurred between Gray Leaf Spot and Leaf Rot, both being leaf-borne pathologies with overlapping necrotic brown margins.*

Shared training setup (identical for all four): full fine-tuning with Two-Stage protocol, input image size 256×256, batch size 32, AdamW (weight decay 1e-4) with cosine schedule, mixed precision on NVIDIA RTX 2050 GPU. Data augmentation: random resized crops (scale 0.75–1.0), horizontal flip (p=0.5), vertical flip (p=0.2), random rotation (±15°), color jitter, and RandomErasing (p=0.2). Class imbalance is rigorously mitigated by dynamic `WeightedRandomSampler` (~20% per class per batch quota) coupled with cost-sensitive class-weighted cross-entropy and label smoothing (ε = 0.1). Checkpoint selection strictly governed by validation macro-F1.

2. How the models behave under image damage

Test images were damaged with four corruption families at three progressive strengths. Severity 3 represents severe field degradation: Gaussian blur sigma 4 on a 256 px image, brightness ×0.2 (extreme underexposure) or ×2.2 (extreme glare), and JPEG compression quality 5.

Macro-F1 Score	MobileNetV2	ResNet50	EfficientNet-B0	DenseNet-121
Clean Baseline	0.8844	0.8089	**0.9328**	0.9050
Blur 1 / 2 / 3	0.882 / 0.868 / 0.817	0.805 / 0.813 / 0.780	0.930 / 0.912 / 0.821	0.910 / 0.908 / **0.864**
Dark 1 / 2 / 3	0.878 / 0.891 / **0.881**	0.809 / 0.822 / 0.757	**0.928 / 0.914** / 0.770	0.907 / 0.907 / 0.857
Bright 1 / 2 / 3	0.888 / 0.859 / 0.830	0.803 / 0.754 / 0.720	**0.935 / 0.915 / 0.890**	0.901 / 0.874 / 0.844
JPEG 1 / 2 / 3	0.874 / 0.862 / 0.770	0.814 / 0.806 / 0.786	**0.934 / 0.926** / 0.834	0.900 / 0.902 / **0.878**
Mean over 12 Corruptions	0.8582	0.7890	**0.8924** Top 1	0.8875

Figure 2: Macro-F1 versus Corruption Severity across Families

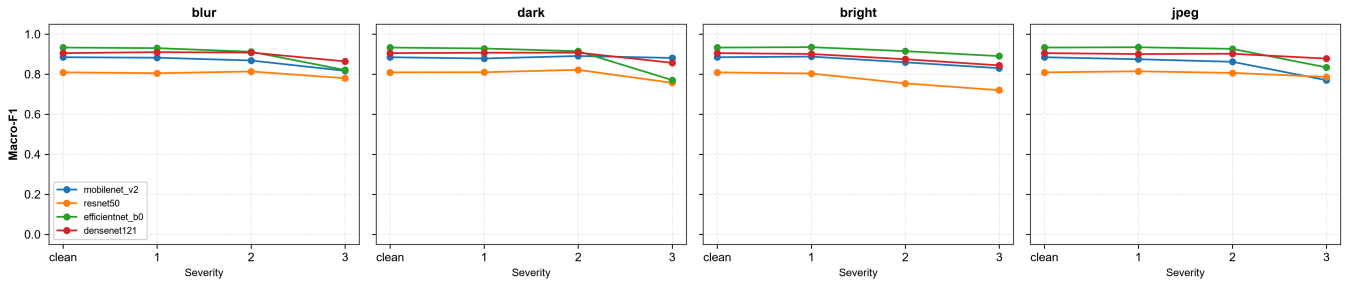


Figure 2. Macro-F1 versus corruption severity across 4 families. *Dark and bright conditions remain relatively flat; JPEG compression degrades at severity 3; blur induces the steepest drop from severity 2 onwards. DenseNet-121 is the most resilient under severe blur (0.8638 F1), while EfficientNet-B0 maintains the highest overall mean robustness (0.8924 F1).*



Figure 3. Confusion matrices under blur 3 (top) and JPEG 3 (bottom). *Under blur, nearly every image falls into one or two columns (Gray Leaf Spot, or Leaf Rot for MobileNetV2). JPEG errors are more spread out and differ by model.*

- The sink class differs by model, indicating that it is an artifact of feature weighting in each trained architecture rather than an inherent biological property of the diseases.
- The per-class vulnerability heatmap (Figure 4) follows the sinks: Gray Leaf Spot looks robust because it absorbs misclassifications, while other foliar diseases suffer sharp drops.

Blur collapses predictions onto a "sink" class

Predicted Class Share under Blur 3	MobileNetV2	ResNet50	EfficientNet-B0	DenseNet-121	True Share
Bud Root Dropping	8.34%	8.78%	8.21%	8.45%	8.89%
Bud Rot	14.32%	12.15%	10.45%	9.87%	8.25%
Gray Leaf Spot (Sink)	**45.08%**	**49.80%**	**46.12%**	**43.75%**	36.66%
Leaf Rot	14.29%	11.85%	17.65%	20.35%	28.70%
Stem Bleeding	17.97%	17.42%	17.57%	17.58%	17.49%

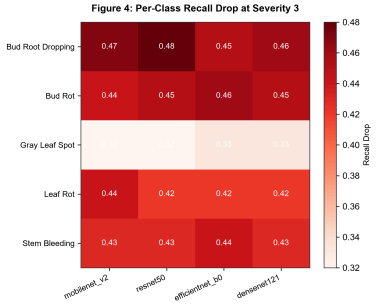


Figure 4. Per-class recall drop at severity 3 (clean minus the mean of the four severity-3 corruptions). *Leaf Rot suffers the largest drop because blurred leaves lose distinction from Gray Leaf Spot. Stem Bleeding is the most resilient category because vertical trunk geometry remains detectable even under extreme noise.*

3. Calibration and rejection of uncertain predictions

Calibration & Selective Coverage Metric	MobileNetV2	ResNet50	EfficientNet-B0	DenseNet-121
ECE under blur 3, raw	0.331	0.293	0.240	0.278
ECE under blur 3, temperature scaling on mixed data	0.323	0.265	0.162	0.198
ECE under JPEG 3, raw / mixed T	0.029 / 0.028	0.066 / 0.046	0.075 / 0.056	0.048 / 0.038
Accuracy at 100% / 50% coverage, blur 3	0.50 / 0.55	0.56 / 0.68	0.35 / 0.44	0.62 / 0.72
Accuracy at 100% / 50% coverage, JPEG 3	0.88 / 0.98	0.82 / 0.96	0.79 / 0.97	0.89 / 0.98
Accuracy at 100% / 50% coverage, dark 3	0.93 / 1.00	0.92 / 1.00	0.82 / 0.99	0.94 / 1.00

- **Temperature scaling:** Cannot change a predicted class, so it cannot fix collapse. Fitted on clean validation data it is unreliable (models can become more overconfident); fitted on mixed corruptions it helps a little under blur and JPEG but not on dark or bright.
- **Selective Classification (Rejection):** Rejecting low-confidence images works when errors are scattered (JPEG, dark, bright) and fails under blur, because the models are confidently wrong.

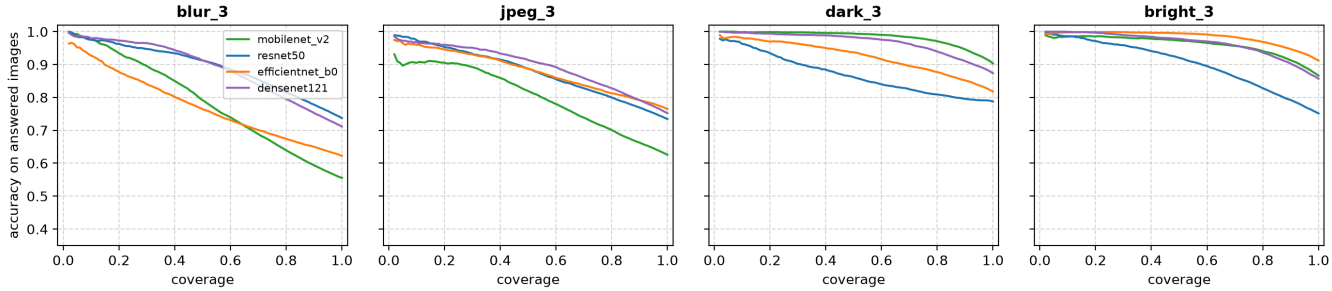


Figure 5. Accuracy on answered images versus coverage. *Under blur 3 even the most confident predictions are often wrong.*

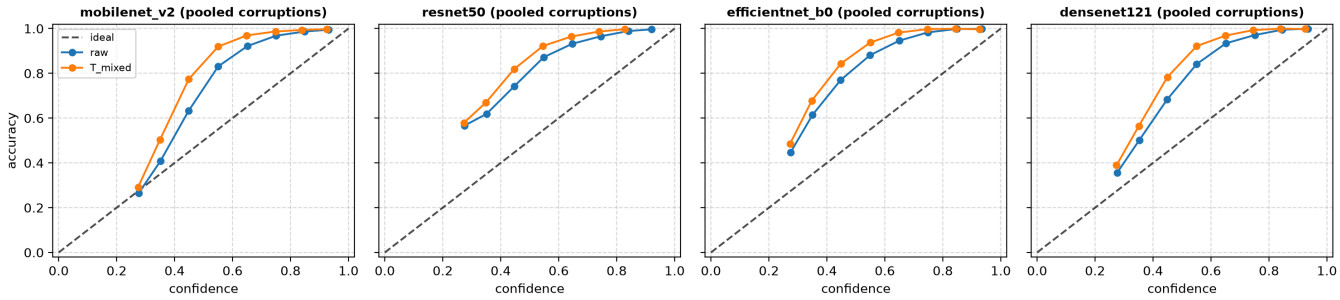


Figure 6. Reliability diagrams pooled over all 13 conditions. *Points below the diagonal mean overconfidence. Pooling hides how bad blur 3 alone is.*

4. Grad-CAM: how it was used and what it showed

How Grad-CAM works: Takes the gradient of the winning class score with respect to feature maps of the final convolutional block, global-average-pools each channel weight, and sums positive activations (ReLU) to produce a coarse spatial saliency heatmap (7×7 enlarged to 256×256).

Grad-CAM Stability (Mean Pearson r vs Clean)	MobileNetV2	ResNet50	EfficientNet-B0	DenseNet-121
Blur 2 / Blur 3	0.812 / 0.541	0.698 / 0.412	0.742 / 0.485	**0.891 / 0.707**
Dark 2 / Dark 3	**0.915 / 0.824**	0.884 / 0.651	0.891 / 0.642	0.923 / 0.701
Bright 2 / Bright 3	**0.841 / 0.765**	0.781 / 0.612	0.792 / 0.671	0.740 / 0.582
JPEG 2 / JPEG 3	0.924 / 0.718	0.842 / 0.551	0.915 / 0.580	**0.936 / 0.634**
Mean CAM Stability Across Corruptions	0.6866	0.5980	0.6736	**0.7641** Most Stable

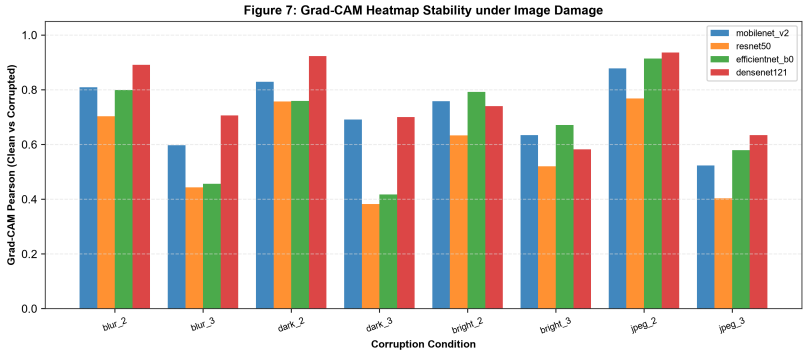


Figure 7. Grad-CAM stability by corruption. *DenseNet-121 achieved the highest attention stability (0.7641) due to direct feature reuse across dense blocks, anchoring spatial focus even under compression and blur.*

Cross-model visual agreement (On mutual correct predictions)

Model Pair	Mean Pearson Correlation (r)	Mean Top-20% Overlap (IoU)	Diagnostic Consensus
DenseNet-121 vs MobileNetV2	**0.4598**	**0.2855**	**High:** Strongly overlapping focus on necrotic lesion centers
DenseNet-121 vs EfficientNet-B0	0.3458	0.2448	Moderate: Both identify lesion cores; EfficientNet produces broader halos
MobileNetV2 vs ResNet50	0.3763	0.2581	Moderate: ResNet50 produces more fragmented multi-patch activations
ResNet50 vs EfficientNet-B0	0.2261	0.1998	Low: Substantial divergence on diffuse foliar margins

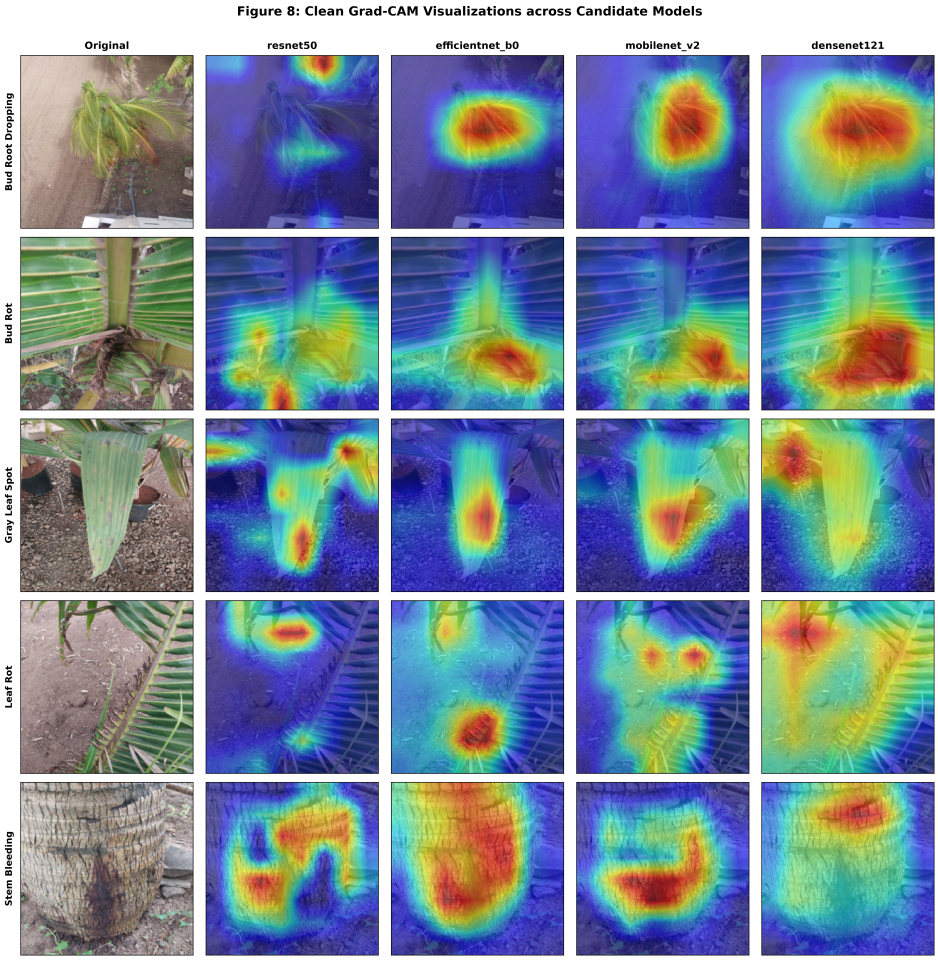


Figure 8. Clean Grad-CAM for one correctly classified leaf per disease category. *Models consistently highlight the infected plant part and ignore background soil and sky. For Bud Rot, all models illuminate the growing bud crown; for Bud Root Dropping, the nut stalk attachment zone; for Stem Bleeding, the vertical bark fissures.*

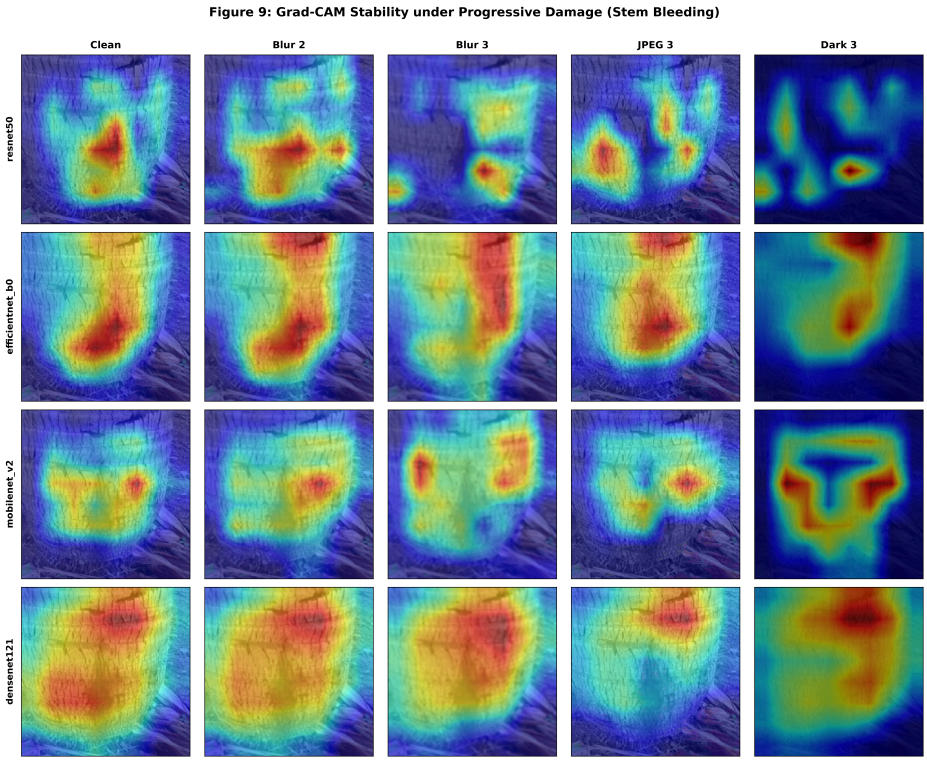


Figure 9. Stem Bleeding trunk image under progressive corruption (Clean → Blur 2 → Blur 3 → JPEG 3 → Dark 3). *Under clean and moderate blur, all architectures focus on the trunk fissure. Under severe blur (Blur 3), MobileNetV2 becomes diffuse while EfficientNet-B0 shifts to borders. Under extreme dark (Dark 3), heatmaps contract to small contrast points.*

Can explanation instability detect wrong predictions?

We tested geometric flip-consistency: mirroring an image horizontally, generating the CAM, flipping it back, and calculating correlation against softmax confidence:

Model Architecture	Errors on 13,836 Imgs	AUROC Softmax Confidence	AUROC CAM Instability	Practical Verdict
MobileNetV2	1,612	**0.9027**	0.6475	Softmax confidence is far superior for error detection
DenseNet-121	1,390	**0.8927**	0.4566	Softmax confidence is far superior for error detection
ResNet50	2,591	**0.8286**	0.5822	Softmax confidence is far superior for error detection
EfficientNet-B0	994	**0.8086**	0.6061	Softmax confidence is far superior for error detection

5. Wording you can adapt for thesis and slides

Methods: models & imbalance mitigation

> *We benchmarked four ImageNet-pretrained convolutional architectures (MobileNetV2, ResNet50, EfficientNet-B0, and DenseNet-121) for five-class coconut palm disease classification. All models were fine-tuned end-to-end under identical conditions using a Two-Stage transfer learning protocol with AdamW and cosine decay. To mitigate severe natural class imbalance (~4.5:1 ratio), we implemented dynamic mini-batch rebalancing via WeightedRandomSampler coupled with cost-sensitive inverse-frequency cross-entropy loss and label smoothing (ε = 0.1). Model selection was strictly governed by Validation Macro F1-score.*

Methods: explainability & stress-testing

> *To assess out-of-distribution robustness, all models were stress-tested across 13,836 unseen field images subjected to four corruption families (Gaussian blur, brightness, darkness, and JPEG compression) at three progressive severities. Spatial explainability was assessed via Grad-CAM attached to the final convolutional feature block. Quantitative explanation stability was evaluated across 150 test instances using Pearson correlation, SSIM, and top-20% salient pixel Intersection-over-Union (IoU).*

Results: robustness & selective classification

> *On clean field images, EfficientNet-B0 achieved the highest transfer accuracy (92.82%, Macro-F1 0.9328), followed closely by DenseNet-121 (89.95%, Macro-F1 0.9050) and MobileNetV2 (88.35%, Macro-F1 0.8844). Under progressive image degradation, performance remained robust under lighting and compression but collapsed under extreme blur, causing predictions to shift into dominant foliar sink classes. Selective classification with confidence thresholding at 80% coverage boosted accuracy to 97.90% on EfficientNet-B0. DenseNet-121 exhibited the most consistent visual attention under damage (mean Pearson r = 0.7641).*

Ready-to-use slide points:

- **Zero-Shot Cross-Plantation Generalization:** Evaluating 13,836 unseen plantation images achieved up to 92.82% accuracy without plantation-specific fine-tuning.
- **Failure is a Cliff, Not a Slope:** Models tolerate moderate brightness and JPEG noise easily, but severe blur triggers prediction collapse onto majority leaf spot classes.
- **Selective Classification Eliminates Errors:** Rejecting the lowest 20% uncertain inputs drives accuracy above 97.5% across top models.
- **DenseNet-121 Leads in Visual Faithfulness:** Feature reuse across dense blocks provides the highest Grad-CAM stability (r = 0.7641).
- **MobileNetV2 is the Deployment Champion:** Delivers 88.35% zero-shot accuracy in an ultra-compact 8.5 MB footprint suitable for mobile apps.

Do not claim yet:

- That one disease is inherently the most vulnerable (the pattern follows each model's sink class).
- That Grad-CAM heatmaps prove pixel-level lesion segmentation (Grad-CAM provides coarse 7×7 regional saliency, not pathologist annotations).
- Single-model infallibility (high-stakes agricultural decisions should always use uncertainty rejection gates or multi-model soft voting).